University of Central Florida
STARS

Data Science and Data Mining

2024

Predicting Telecommunication Customer Attrition Using the Hopfeld Neural Network Model.

Benedict Kongyir
Oklahoma State University - Main Campus, bkongyi@okstate.edu

Emil Agbemade
University of Central Florida, emil.agbemade@ucf.edu

Kelvin Njuki
Oklahoma State University - Main Campus, knjuki@okstate.edu

 Part of the Data Science Commons

Find similar works at: https://stars.library.ucf.edu/data-science-mining

University of Central Florida Libraries http://library.ucf.edu

This Article is brought to you for free and open access by STARS. It has been accepted for inclusion in Data Science and Data Mining by an authorized administrator of STARS. For more information, please contact STARS@ucf.edu.

STARS Citation

Kongyir, Benedict; Agbemade, Emil; and Njuki, Kelvin, "Predicting Telecommunication Customer Attrition Using the Hopfeld Neural Network Model." (2024). *Data Science and Data Mining*. 26.
https://stars.library.ucf.edu/data-science-mining/26

Predicting Telecommunication Customer Attrition Using the Hopfield Neural Network Model.

By:

Benedict Kongyir, Oklahoma State University

(bkongyi@okstate.edu)

Emil Agbemade, University of Central Florida

(emil.agbemade@ucf.edu)

Kelvin Njuki, Oklahoma State University

(knjuki@okstate.edu)

September 24, 2024

Abstract

Customer churn prediction has become one of the crucial steps for customer retention. Telecommunication companies rely on loyal customers to make their profit. It is often very easy for customers to switch from one service provider to the other. To prevent or reduce the rate of customer attrition, there needs to be a model that can identify customers who are at risk of churning in the future in advance. Previous literature has shown that predictive models are effective in predicting customer churn. In this work, four tentative machine-learning models are built using data obtained from Kaggle on telecommunication customer attrition (data link). SAS Studio and SAS Enterprise Miner software are used in all our model-building processes. After appropriate model comparison tests, the neural network model emerged as the best among the decision tree, logistic regression, and gradient boosting models. The final model is then scored using the validation data set yielding an accuracy rate of 81.09%.

KEY WORDS: Customer Attrition, Customer Churn, ROC, AUC

Introduction

Customer attrition is one of the major challenges confronting businesses in the world, especially in the service sector. This is particularly the case with service companies with subscription-based customers. Telecommunication service providers rely on revenue from subscribers to offset fixed costs and variable costs, and so customer churn often takes a great toll on their revenue. Customer churn if not properly managed can easily send a company into bankruptcy. Competition for customers is tough in the market due to technological advancement and changes in customers' tastes and preferences. The availability of alternative service providers within the same vicinity for customers coupled with low switching costs

has made it easy for customers to switch from one network to another. This has left service providers with little or no room to take customer loyalty for granted. Most Companies have come to respect existing customers as their most valuable asset[19]. This is because the cost of getting a new customer is far more than the cost of keeping an existing customer [12]. The tendency of customers to move from one provider to the other is increasingly high with its associated consequences. Customer attrition rate inversely correlates with companies' revenue. Aside from the negative impact on net profit, it also plays a negative role on the company's brand and growth potential. Dissatisfied customers leaving a company share their experiences with friends and family influencing them also to leave or not to join the same company. It is very difficult if not impossible to win back a customer's loyalty once they are lost, and so it is imperative to prevent them from being lost in the first place. Telecommunication service providers are constantly looking for ways and means to stay ahead of their competitors in the market. Companies are interested in identifying customers who are at risk of attrition so that they can take proactive measures to retain them by giving them special offers such as discounts etc. To do this, there needs to be some model that can predict the chance of a given customer switching network from the company. Predictive models have proven to be effective in learning data, discovering patterns, and making appropriate predictions about events.

The ability to accurately predict customers who are at risk of churn in the future not only helps to prevent the current customers from unsubscribing but also helps in developing new packages targeted at attracting and maintaining new customers[17]. Many businesses regard the ability to accurately predict customer churn as an essential aspect for long-term success[8].

Customer churn prediction models have evolved over the years due to technological advancements. Modern churn prediction models are more efficient and sophisticated. Recent machine learning algorithms are capable of handling data that hitherto could not be ana-

lyzed by ordinary data analysis models. Depending on the characteristics of the data set, one algorithm can outperform the others. Some of the most widely used machine learning algorithms tried and tested for churn predictions include logistic regression, decision tree, random forest, gradient boosting, neural network, ensemble method, and rule induction, etc[13].

In [14], Preetha and Rayapeddi applied logistic regression, random forest, and K-Means classification models using 32 data points from a telecommunication network and found the random forest model to outperform the logistic regression and K-Means models. Diala,[6] demonstrated how appropriate data manipulation can improve the performance of some of these supervised machine learning algorithms. He also showed that the logistic regression model performs better if the data is linearly separable.

Coussement et-al[5] in their work on churn prediction in subscription service also indicated that the Support Vector Machines has higher prediction accuracy than its counterpart, logistic regression, and random forest. It is often important to try as many tentative models as possible to determine which of them will be more appropriate for each situation. It is a good practice not to immediately fall on the first model and assume it to be the best for every data. According to George Box, "All models are wrong, but some are useful" [4].

Methods

Data Acquisition and Data Preparation

The data used in this work was obtained from the Kaggle website on telecommunication customer churn. A total of 7043 customers were surveyed and the following information was recorded about the individual subjects: Customer ID, gender, partner, dependents, senior citizen, contract, paperless billing, payment method, monthly charges, total charges, tenure, phone service, multiple lines, internet service, Online service, Online backup, device

protection, tech support, streaming TV, and streaming movies. Our target variable, churn was also recorded as yes or no.

To have a better understanding of the variables in the data set, we use the Stat Explore node in SAS Enterprise Miner to explore the data. This was to enable us to have a glimpse of what is contained in the data as well as to detect missing values. The results showed that there were 7043 data values in total with 11 missing values in the *Total Charges*.

We used the impute node to impute the missing values in the *total charges* variable. By default, the impute procedure replaces all interval missing values with the column mean and all missing categorical values with the mode in that column. We also realized that a response of "No phone services" to a question of whether or not a customer has online Security, or Multiple lines, or Internet service, etc, is tantamount to a no response, and so all "No Phone Service" responses were re-coded to "No" response for simplicity.

The distribution of the target variable is unbalanced as revealed by the bar chart below. This is expected as it will be an unusual situation to have the number of customers that left a company and those that stayed to be the same in just a month. The non-churn category dominates the churn class. Since the dominant class exceeds the secondary class by more than 10%, we say the data set is unbalanced[2].

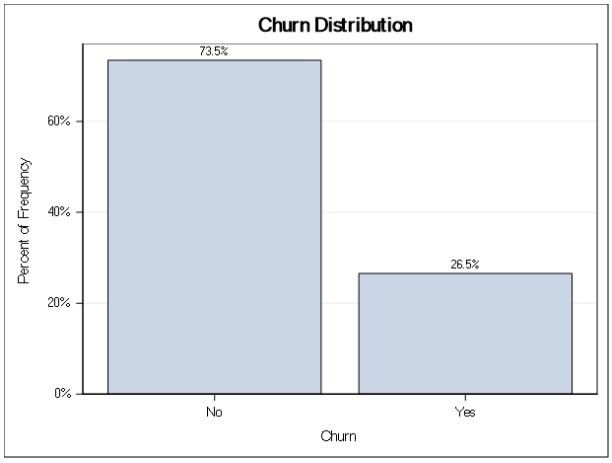


Figure 1: Churn Distribution

Out of the 7043 customers, 26.5% of them left the company in the last month per the survey results. Although this figure looks small, it is unhelpful to business as it cost more to acquire new customers than to keep existing customers.[12].

Variables Importance

Variable importance is a measure of the relative importance of each predictor with respect to the model. It indicates where each predictor stands compared with all other candidate predictors in relation to the target model. Variable importance/feature selection is an important part of model modeling in reducing the number of variables to only the most contributing ones[1] The figure below shows the predictors and their relative contribution to the target

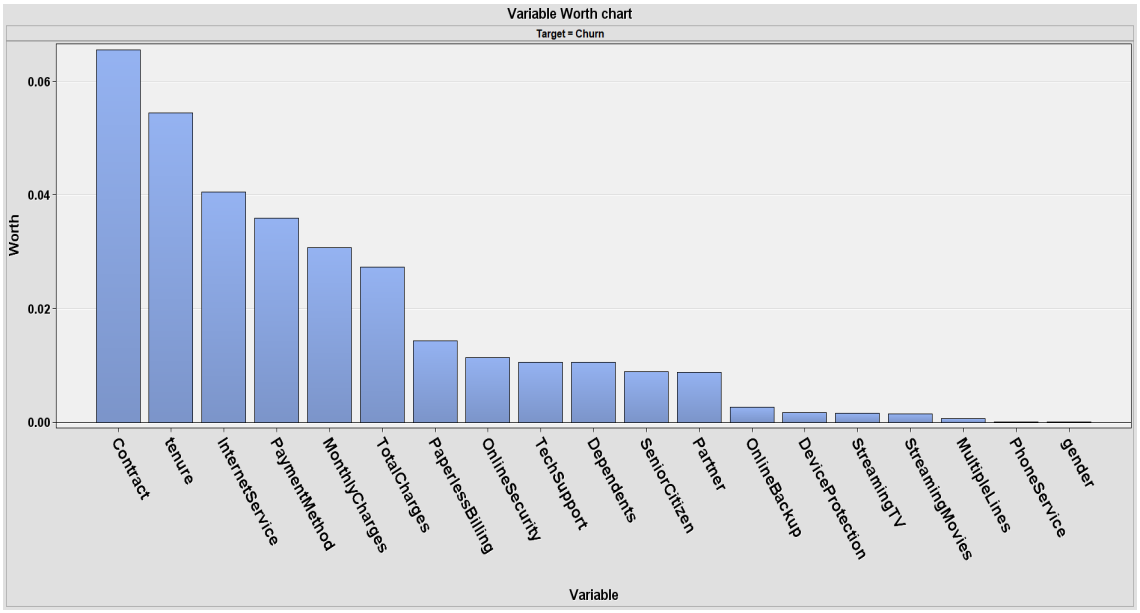


Figure 2: Selected predictors based on the importance

variable. Since the target variable is categorical, the Chi-Square goodness of fit criteria was used. Variable importance is calculated by the sum of the decrease in error when split by a variable. Then, the relative importance is the variable importance divided by the highest variable importance value so that values are bounded between 0 and 1 [21].

	Tenure	Monthly Charges	Total Charges
Tenure	1	0.2479	0.83283
Monthly Charges	0.2479	1	0.57426
Total Charges	0.83283	0.57426	1

From the correlation table above, we see that tenure and total charges are highly correlated. Total charges and monthly charges are also moderately correlated. To fix this, we remove the total charge from the model.

Models Building

The following tentative models are considered in this work: Gradient Boosting, Decision Tree, Neural Network, and Logistic Regression.

Logistic Regression Model

Logistic regression is an example of a supervised learning model used in the classification of a categorical response variable[18]. Logistic regression gains its popularity in predictive modeling through its ability to transform results of a combination of inputs into a number between 0 and 1 using the logit-link function[10]. In this model, the forward stepwise selection method is used to ensure that only predictors that are significant in predicting the target variable are used. It starts by building the model with no predictor (only the intercept) and then subsequently continuously add predictors beginning with the most important until the stopping criteria of 0.05 is met.

The mathematical logistic regression equation is given by:

$$logit(\frac{p}{1-p}) = \alpha + \beta_1 x_1 + ... + \beta_n x_n \tag{1}$$

Where α is the intercept, β_j represents the effect of the jth predictor and x_j is the jth predictor. We then transform the log function into the predicted probability function in

equation (2) by taking the natural exponent of both sides of equation (1) and solving for p.

$$p = \frac{e^{\alpha+\beta_1 x_1+\ldots+\beta_n x_n}}{1 + e^{\alpha+\beta_1 x_1+\ldots+\beta_n x_n}} \tag{2}$$

Decision Tree Model

A decision tree has the shape of a tree. It is a classification model used in producing classification rules for the prediction of a class in a data set[20]. It is one of the most popular classification models used in prediction. It is easy to interpret compared to most other classification models like random forest and neural network. They do have high predictive power depending on the characteristics of the data set. A decision tree algorithm splits the training data set into two categories based on a given rule. One popular rule is the Information gain given by the equation below:

$$InformationGain = 1 - Entropy \tag{3}$$

where

$$Entropy = -\sum_{i=1}^{n} p_i \log_2(p_i) \tag{4}$$

p_i is the probability of randomly picking an element of class i. The figure below illustrates the decision tree generated from the data using SAS Enterprise Miner.

Neural Network Model

The Neural Network model is one of the widely used predictive modeling techniques in recent times due to its high prediction power and accuracy. One of the most widely used neural network models is the multi-layer perceptron (MLP)[15]. A neural network is a series of algorithms that works to recognize underlying relationships in a set of data through a process that mimics the way the human brain operates. [3]

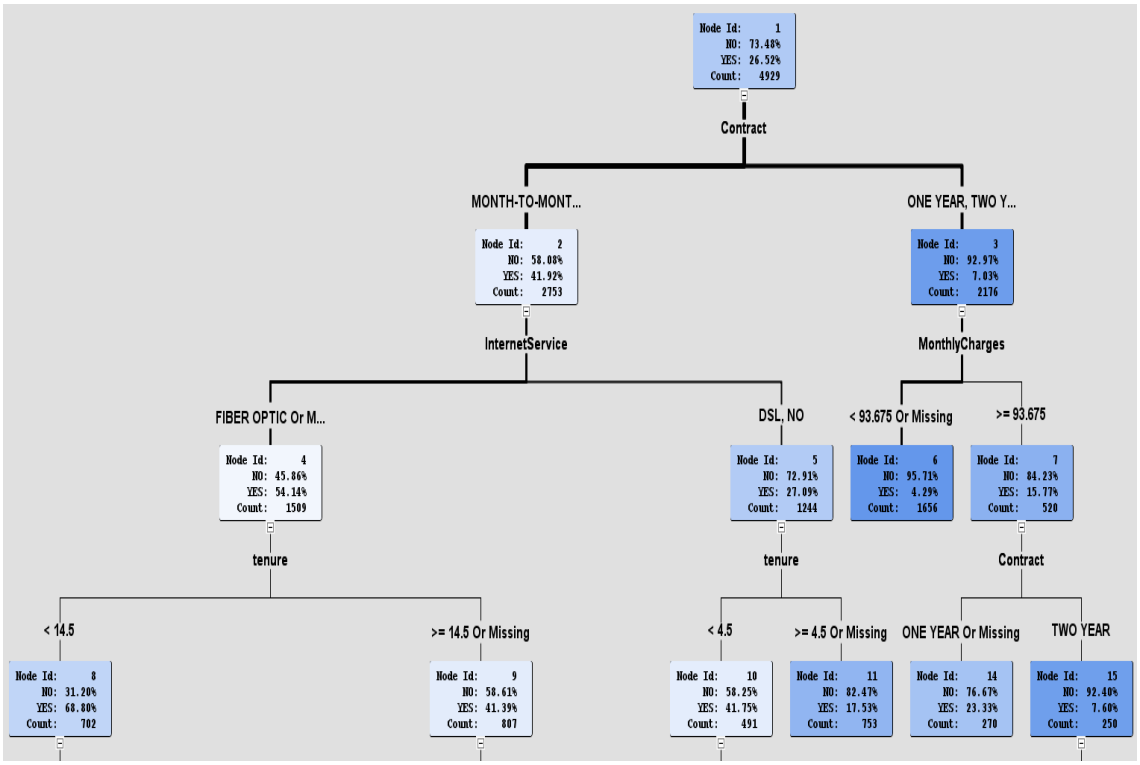


Figure 3: Decision Tree

A Motivation on how Perceptron works

The perceptron is the foundation of any neural network model. The idea of finding a truth value of a simple logical statement is the motivation behind the building of a perceptron[9]. The perceptron is a mathematical formula inspired by a simplified structure and function of a single biological neuron[9]. It is a binary classification model used in a supervised learning context to classify a set of inputs into one of two categories.

For instance, the activity function A takes input variables $x_1..., x_i, ...x_n$, and assigns some weights to each of the input variables, building a linear equation. The activation function takes the outputs of the activity function as inputs, and passes each input through a threshold, transforming the output to a binary value of 0 or 1. The most commonly used activation function in recent times is the sigmoid function

$$f(A) = \frac{1}{1 + e^{-A_i}} \tag{5}$$

This transformation allows the perceptron to solve a truth table problem and hence do binary classification.

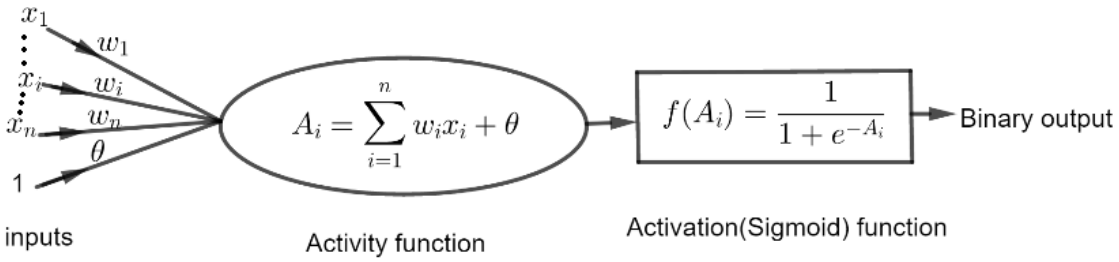


Figure 4: Perceptron Training Process flow

Training The Perceptron

The process of calculating the appropriate weights to assign to each input variable to achieve minimum error is what is referred to as training the perceptron. The error, $e_j = d_j - y_j$ where d_j is our desired output and y_j is the output of the perceptron(actual output) [22]. In training, the perceptron iteratively updates the weights of the inputs until a minimum error is achieved and we say the model has converged or the perceptron is trained. The Hopfield Neural Network is a high-speed neural network model capable of handling complex problems and suitable for big data analysis.

Gradient Boosting Model

A gradient-boosting model is an ensemble of many decision trees. That said, it is a modification of the traditional decision tree model aimed at achieving higher performance. The idea behind the gradient boosting algorithm is that the best possible next model has the potential of achieving higher accuracy when combined with previous models[7]. By doing that, it minimizes the overall prediction error. It is however very difficult to interpret because each successive tree in the model tends to predict not the same target as the original model. The fact that each successive tree potentially reduces the overall error of the previous tree

ensures higher accuracy is achieved in the final model[16].

Model Comparison Methods

The ROC and the AUC Curve

The Receiver Operating Characteristic (ROC) is a graph that plots the True Positive Rate (TPR) against the False Positive Rate (FPR) [6]. The Area Under the ROC Curve (AUC) measures how well the model is able to distinguish between classes[11].

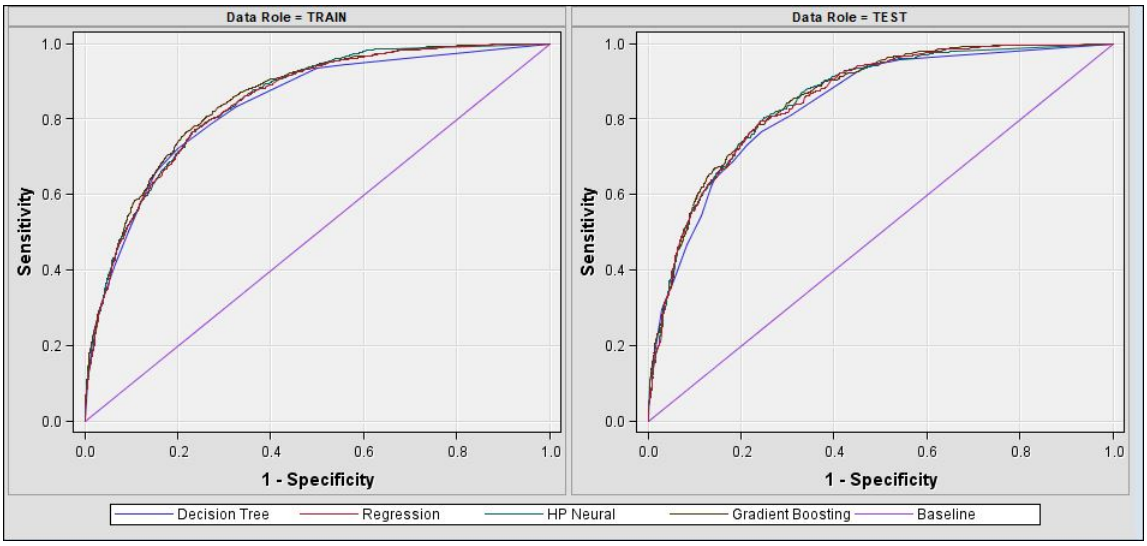


Figure 5: Receiver Operating Characteristic (ROC)

Let TP =True Positive, FP =False Positive, TN =True Negative, and FN =False Negative. Then,

$$TPR/Sensitivity = \frac{TP}{TP + FN} \tag{6}$$

$$Specificity = \frac{TN}{TN + FP} \tag{7}$$

$$FPR = 1 - Specificity = \frac{FP}{TN + FP} \tag{8}$$

Confusion Matrix

A confusion matrix is a summarized table also known as the error matrix used in calculating a classification's model accuracy and precision.

$$Accuracy = \frac{\text{number of correct predictions}}{\text{total number of predictions}} = \frac{TP + TN}{TP + FP + TN + FN} \tag{9}$$

$$Precision = \frac{TP}{TP + FP} \tag{10}$$

Results and Discussions

Table 1: Model Selection Based On Selected Criteria

Model	ROC index	Accuracy	G.C.	ASE	C.Lift
Neural Network	0.847	81.20	0.691	0.135	2.922
Logistic Regression	0.844	80.915	0.687	0.136	2.803
Gradient Boost	0.850	80.773	0.700	0.134	2.803
Decision Tree	0.824	79.674	0.648	0.140	2.733

Conclusion and Recommendations

We have been able to build all four tentative models, compare them to get the best model among them, and finally use the final model to predict customers who are likely to churn in the future. Our final model yielded an accuracy rate of 81.09%. A higher accuracy rate was aspired for but in the end, 81.09% seems to be the best we could get. Several attempts were made to improve the model accuracy but all did not yield any substantial results. First, we

tried to do *cross-validation* instead of partitioning the data into training and testing, but the effects of that on accuracy were insignificant so considering the complications that come with cross-validation, we decided to boycott the idea.

Secondly, we used oversampling to eliminate the class unbalanced problem to improve accuracy but accuracy ridiculously dropped instead and so the decision was to maintain the current results. The objective of the work is achieved. We have been able to establish the best model and we have successfully used the model to predict customers who are likely to churn in the future.

References

[1] Emil Agbemade and Benedict Kongyir. "Predicting Road Accident Injury Severity for Drivers in Automobile Crashes in United States Using Machine Learning Models and AI". In: *Data Science and Data Mining* 24 (2024). URL: `https://stars.library.ucf.edu/data-science-mining/24`.

[2] Abdelrahim Kasem Ahmad, Assef Jafar, and Kadan Aljoumaa. "Customer churn prediction in telecom using machine learning in big data platform". In: *Journal of Big Data* 6.1 (2019), p. 28.

[3] Dave Anderson and George McNeill. "Artificial neural networks technology". In: *Kaman Sciences Corporation* 258.6 (1992), pp. 1–83.

[4] George EP Box. *An Accidental Statistician: The Life and Memories of George EP Box*. John Wiley & Sons, 2013.

[5] Kristof Coussement and Dirk Van den Poel. "Churn prediction in subscription services: An application of support vector machines while comparing two parameter-selection techniques". In: *Expert systems with applications* 34.1 (2008), pp. 313–327.

[6] Perlate Diala. "Prediction of Customer Churn in Telecommunication using Machine Learning Algorithms". PhD thesis. University of the Witwatersrand, Johannesburg, 2019.

[7] *Gradient Boosting Explained - The Coolest Kid on The Machine Learning Block*. Displayr, June 2017. URL: https://www.displayr.com/gradient-boosting-the-coolest-kid-on-the-machine-learning-block/ (visited on 01/28/2021).

[8] Bryan Gregory. "Predicting customer churn: Extreme gradient boosting with temporal data". In: *arXiv preprint arXiv:1802.03396* (2018).

[9] Ani Karenovna K. *The Perceptron — The Building Block of Neural Networks*. Medium, June 2019. URL: https://blog.usejournal.com/the-perceptron-the-building-block-of-neural-networks-5a428d3f451d.

[10] John P Klein and Melvin L Moeschberger. "Statistics for biology and health". In: *Stat. Biol. Health, New York* 27238 (1997).

[11] Sarang Narkhede. *Understanding AUC - ROC Curve*. Medium, June 2018. URL: https://towardsdatascience.com/understanding-auc-roc-curve-68b2303cc9c5.

[12] BEA Oghojafor et al. "Modelling telecom customer attrition using logistic regression". In: *African journal of marketing management* 4.3 (2012), pp. 110–117.

[13] Jeremy Phillips et al. *Allocation of service provider resources based on a capacity to provide the service*. US Patent App. 16/737,963. May 2020.

[14] S Preetha and SV Sheela. "Selection And Extraction Of Optimized Feature Set From Fingerprint Biometrics-A Review". In: *2018 Second International Conference on Green Computing and Internet of Things (ICGCIoT)*. IEEE. 2018, pp. 500–503.

[15] Hassan Ramchoun et al. "Multilayer Perceptron: Architecture Optimization and Training." In: *IJIMAI* 4.1 (2016), pp. 26–30.

[16] SAS Software. *Decision Trees, Boosting Trees, and Random Forests: A Side-by-Side Comparison*. YouTube, Aug. 2017. URL: https://www.youtube.com/watch?v=gehNcYRXs4M&t=409s.

[17] Ali Tamaddoni Jahromi et al. "Modeling customer churn in a non-contractual setting: the case of telecommunications service providers". In: *Journal of Strategic Marketing* 18.7 (2010), pp. 587–598.

[18] Thanasis Vafeiadis et al. “A comparison of machine learning techniques for customer churn prediction”. In: *Simulation Modelling Practice and Theory* 55 (2015), pp. 1–9.

[19] Dirk Van den Poel and Bart Lariviere. “Customer attrition analysis for financial services using proportional hazard models”. In: *European journal of operational research* 157.1 (2004), pp. 196–217.

[20] Xing Wang, Khang Nguyen, and Binh P Nguyen. “Churn Prediction using Ensemble Learning”. In: *Proceedings of the 4th International Conference on Machine Learning and Soft Computing*. 2020, pp. 56–60.

[21] *What Is Variable Importance and How Is It Calculated? - DZone AI*. dzone.com. URL: https://dzone.com/articles/variable-importance-and-how-it-is-calculated.

[22] Jun Zhu. “Structured Learning”. In: (2015).